Miles Zhou

Presenter · slides 6–13 · about 7:47 of narration

> THE PROJECT STORY
>
> We wanted to study dispatch decisions. April's data described aggregate patterns rather than individual journeys, so we built plausible evenings from evidence and explicit assumptions. We checked the implementation, replayed identical evenings under different transfer rules, and found a conditional trade-off between timely service and Lyft use. Our next work tests how robust those findings are.

Contents

1 Your section 1

2 Essential preparation reading 2

3 Concepts you must explain confidently 2

4 Questions you lead on 2

5 Your narration (exact words) 4

6 Manual verification activity: The no-show denominator 8

7 Rehearsal and readiness checklist 11

1 Your section

Slides 6–13: April describes totals—not individual journeys → The evidence becomes a scenario blueprint.

Purpose: Aggregate evidence, workload, scenario construction and assumptions.

You start immediately after the handoff sentence: "That gives us the question. Miles will now explain how April's aggregate data became the ingredients of our scenarios." Begin with slide 6's first sentence; do not re-introduce yourself.

Your handoff sentence (end of slide 13): "Those inputs define the constructed evening. Youzhi will explain how the model makes decisions as that evening unfolds."

2 Essential preparation reading

Read these in the shared guide (`01_Team_Presentation_Guide.pdf`, one folder up) and the methodology PDF:

- Slide 6 · April describes totals—not individual journeys: guide, slide 6 (guide p. 20); methodology §2.1 What the dashboard gives, p. 10; §2.2 Why aggregates cannot pin down one scenario, p. 11
- Slide 7 · How busy should a constructed evening be?: guide, slide 7 (guide p. 21); methodology §3.1 Volume: how many requests, p. 12
- Slide 8 · We started with a smaller, deliberately demanding hour: guide, slide 8 (guide p. 23); methodology §7.7 M3: the L5 pilot and the 30-night batch, p. 40; §4.12 Designed checks: the level ladder, p. 25
- Slide 9 · Knowing the endpoints does not identify the journeys: guide, slide 9 (guide p. 24); methodology §3.3 Geography, p. 14; §3.4 Origin–destination pairing, p. 15
- Slide 10 · One average can hide very different groups: guide, slide 10 (guide p. 25); methodology §3.5 Parties: how many riders per booking, p. 15
- Slide 11 · One status label hides different events: guide, slide 11 (guide p. 26); methodology §3.7 Patience: when a waiting rider gives up, p. 16
- Slide 12 · The simulator needs a probability at vehicle arrival: guide, slide 12 (guide p. 27); methodology §3.8 No-shows: who is at the curb, p. 17
- Slide 13 · The evidence becomes a scenario blueprint: guide, slide 13 (guide p. 28); methodology §3.10 Patterns, seeds and pairing, p. 19; §1.2 Three things kept apart, p. 6
- Everyone: guide sections “What every member must be able to explain” and “Quick Q&A navigation”.

3 Concepts you must explain confidently

- observed vs derived vs assumed vs designed
- the workload chain and its two stated assumptions; 168 is planned, not run
- the two-van pilot: 28 requests, the 18:20–18:30 burst, 30 nights $\times$ 6 rules
- why place totals do not identify trips; why one average does not fix a distribution
- “Canceled” as an observation, an assumed mapping and hidden patience; the two no-show denominators

4 Questions you lead on

Answer these first; route anything else to its owner (guide, Quick Q&A navigation).

- What April’s data do and do not contain — guide slides guide, slide 6 (guide p. 20); backups 33
- Where 168 requests and 12.85 per vehicle come from — guide slides guide, slide 7 (guide p. 21); backups 33
- The two-van pilot and the burst — guide slides guide, slide 8 (guide p. 23); backups 33, 35
- Trip pairing, party sizes, patience, no-shows — guide slides guide, slide 9 (guide p. 24), guide, slide 10 (guide p. 25), guide, slide 11 (guide p. 26), guide, slide 12 (guide p. 27); backups 33
- What is observed, derived or assumed — guide slides guide, slide 6 (guide p. 20), guide, slide

13 (guide p. 28); backups 33

5 Your narration (exact words)

Slide 6 · April describes totals—not individual journeys

about 58 s · open slide

SAY (exact narration)

We know how many requests ended in each status, where pickups and drop-offs accumulated, and how activity varied by hour. We cannot reconstruct each rider's journey. [optional] On the left is what April gives; on the right, what it never gives: when a rider asked, where they went, how long they waited, and what happened next. Throughout the talk we label inputs. Observed means read from the dashboards. Derived means computed from observed numbers. Assumed means our choice, to be tested; designed means an experiment setting we chose. We therefore built plausible evenings from the evidence, while keeping the missing pieces visible.

TRANSITION

The first piece is how busy an evening should be.

Slide 7 · How busy should a constructed evening be?

about 64 s · open slide

SAY (exact narration)

We estimated a workload from April's completed rides, request-to-completion ratio and reported vehicle availability. About 5,100 completed rides at six o'clock over thirty days is about 170 in one evening. April had about 2.09 requests for every completed ride, so roughly 355 requests in that hour. Divided by 27.6 vehicles on duty, that is about 12.85 requests per vehicle per hour. This assumes the monthly request-to-completion ratio applies within the hour, and uses vehicle availability reported across services. It is an estimated workload, not observed arrivals. Our planned evening uses five vans, a team design choice; each hour then gets its own count, 64, 50 and 54: 168 requests. That five-van evening is planned, not yet run.

TRANSITION

What we actually ran is smaller, and deliberately harder.

Slide 8 · We started with a smaller, deliberately demanding hour

about 56 s · open slide

SAY (exact narration)

The current results come from two vans and 28 requests in one hour, repeated across 30 constructed nights. Each van has fourteen seats. Requests arrive between six and seven, and eleven of the 28 are placed in a burst between 6:20 and 6:30. Each of the 30 nights is different, and each is run under six transfer rules, defined shortly. This is a controlled pilot, not a reconstruction of a typical April night. Its load, fourteen requests per van-hour, is close to the 12.85 we estimated per vehicle across all services, but the burst concentrates demand on purpose, to stress the transfer decision.

TRANSITION

Next, where those requests go.

Slide 9 · Knowing the endpoints does not identify the journeys

about 49 s · open slide

SAY (exact narration)

Two different sets of trips can produce exactly the same pickup and drop-off totals. In this toy example, both versions have identical green totals, which are all the maps show, yet different trips. So pairing pickups with drop-offs is a modelling choice. We assumed nearby journeys were more likely, then balanced the synthetic trips to preserve the observed location totals. The rate at which distance lowers a pair's weight is our assumption; the data cannot identify it. [optional] The map shows one constructed hour built this way.

TRANSITION

The next input is who travels together.

Slide 10 · One average can hide very different groups

about 61 s · open slide

SAY (exact narration)

April helps identify an average group size. It does not tell us the full distribution of groups. April gives about 1.39 van passengers per completed van request. Examples A and B have exactly that average, yet one has eight percent of requests with three or more riders and the other over twelve. That matters for seats in the van, and because our waiting penalty counts every passenger. Our mix covers all requests, so its mean is slightly higher. Our chosen group-size mix is an assumption informed by the average—not a distribution recovered from April. A party of five that is moved to Lyft rides together in one larger XL vehicle.

TRANSITION

The hardest label on the dashboard is “Cancelled”.

Slide 11 · One status label hides different events

about 62 s · open slide

SAY (exact narration)

‘Cancelled’ does not tell us whether a rider gave up or a dispatcher moved the request to Lyft. Almost half of April’s requests ended “Canceled”. We assume that requests moved to Lyft also close as “Canceled” in the van system, and we keep the share that is riders giving up as a range to test, fifty to seventy-five percent. To let riders give up in the simulator, each rider gets a private deadline the dispatcher cannot see, with a median of eighteen minutes. The patience distribution is provisional. It lets the simulator represent abandonment; it is not a behavioural distribution measured from April. [optional] Many patience distributions could produce the same cancellation share.

TRANSITION

A similar puzzle hides in “No Show”.

Slide 12 · The simulator needs a probability at vehicle arrival

about 55 s · open slide

SAY (exact narration)

The share of all requests marked ‘No Show’ is different from the chance of a no-show when a vehicle actually arrives. 2,359 no-shows out of all 50,297 requests is 4.7 percent. But a no-show can only happen when a vehicle arrives, and most cancelled requests never saw one. Out of van arrivals, completed rides plus no-shows, it is about 8.9 percent. That figure holds only if “No Show” was recorded after a real van arrival, “Completed” counts van trips, and Lyft hand-offs sit in “Canceled”. Our arrival-based probability is provisional because it depends on how the dashboard statuses were recorded.

TRANSITION

Together, these pieces form a scenario.

Slide 13 · The evidence becomes a scenario blueprint

about 62 s · open slide

SAY (exact narration)

A scenario combines evidence-based inputs, explicit assumptions and hidden rider behaviour. Demand: arrival times, locations, trip pairing and party sizes. Resources: vans, their fourteen seats and travel times, which we inflate by twenty percent as a placeholder. Hidden behaviour: patience, presence at the curb and Lyft delay; a transfer takes two minutes to order, then the Lyft takes about eight minutes typically. And experiment settings: the van dispatcher, held fixed, and the transfer rule, which we vary. Many inputs carry two labels, derived and assumed. One point matters for what follows: inputs are fixed before a run, but cancellations, no-shows and transfers are not. They are outcomes of the run, not counts we impose.

HANDOFF: Miles Zhou → Youzhi Zhang

Those inputs define the constructed evening. Youzhi will explain how the model makes decisions as that evening unfolds.

6 Manual verification activity: The no-show denominator

> **WHAT HUMAN VERIFICATION MEANS**
>
> A program can generate and check an answer. Human verification means a teammate independently works through a defined example, records their calculation and compares it with the implementation.

This is a guided preparation activity, not a quiz or an exam. Work through the procedure on your own first and write down your result. Only then turn to the reference answer, which is on a separate page. If your result differs, record the difference rather than adjusting your working to match: a disagreement is useful information. The reference answer was produced with AI assistance from the project files; it does not count as a teammate's verification.

Related slides: Slide 12; also slide 11. **Goal:** Calculate the two no-show rates from the verified April counts, explain why they answer different questions, and name the status-mapping assumptions needed before using the second as a probability at vehicle arrival.

Source inputs

Input (April 2026, whole month)	Count	Provenance
All requests (all statuses)	50,297	Dashboard status totals
Completed	24,095	Dashboard status totals
No Show	2,359	Dashboard status totals
Canceled	23,843	50,297 − 24,095 − 2,359 (the digitized status sums also show 99 "Denied")

Procedure

1. Compute Rate 1 = No Show / all requests. Write it as a decimal and as a percentage (one decimal place).
2. Compute Rate 2 = No Show / (Completed + No Show). Write it the same way.
3. For each rate, write the question it answers in one sentence ("Of all requests, …" / "When a vehicle reaches the curb, …").
4. Explain why most cancelled requests do not belong in the denominator of the second rate.
5. List the status-mapping assumptions that must hold before Rate 2 can be used as the probability that a rider is absent when a vehicle arrives.

Worksheet

YOUR WORKING

Step-by-step working (use the lines, or attach your own sheet)

My result

Record

Status	**Awaiting human completion**
Name	
Date completed	
My result	
Matches the reference? (yes / partly / no)	
Discrepancy, and what I did about it	
What remains unverified	

Reference answer

REFERENCE ANSWER (compare only after finishing your own working)

Rate 1. 2,359 / 50,297 = 0.0469, about **4.7%**: the share of *all requests* that ended as "No Show".

Rate 2. Completed + No Show = 24,095 + 2,359 = 26,454. Then 2,359 / 26,454 = 0.0892, about **8.9%**: the share of *van arrivals at the curb* where the rider was absent, *if* the statuses mean what we assume.

Why they differ. A no-show can only happen when a vehicle reaches the curb. Most "Canceled" requests never had a vehicle arrive (the rider gave up, or the request was moved to Lyft), so including them in the denominator dilutes the rate by about half.

Assumptions needed for Rate 2 (the status mapping):

- "Completed" counts van-completed trips, each with exactly one van arrival;
- "No Show" was recorded only after a real van arrival and the 3-minute curb wait;
- requests handed to Lyft were recorded as "Canceled", not as completed or no-show.

These are set out in the methodology, §3.8, p. 17. None of them is confirmed by the data, so 8.9% is a *conditional derivation*, used provisionally; it is not a verified operating probability. The simulator also applies the same probability to Lyft arrivals, which is a further assumption (D-48).

Comparison checklist

- ☐ Do my two rates match 4.7% and 8.9% to one decimal place?
- ☐ Did I state a different question for each rate?
- ☐ Did I explain why cancelled requests leave the second denominator?
- ☐ Did I list all three mapping assumptions, and say they are unconfirmed?
- ☐ Did I avoid calling 8.9% a measured probability?

Record any disagreement in the table above. Do not change your own working to match the reference.

7 Rehearsal and readiness checklist

- ☐ Read my slides' preparation notes in the shared guide.
- ☐ Read my slides in the final deck (`02_Final_Presentation.pdf`) and know which backup slides I may need.
- ☐ Said my narration aloud with a stopwatch; my section took _____ (estimate 7:47).
- ☐ Practised my opening and handoff sentences with the neighbouring presenter.
- ☐ Can explain every concept listed above without the slides.
- ☐ Completed my verification activity and filled in the record (status changed from "Awaiting human completion" only by me).
- ☐ Took part in one timed full-team rehearsal (total time: _____).